

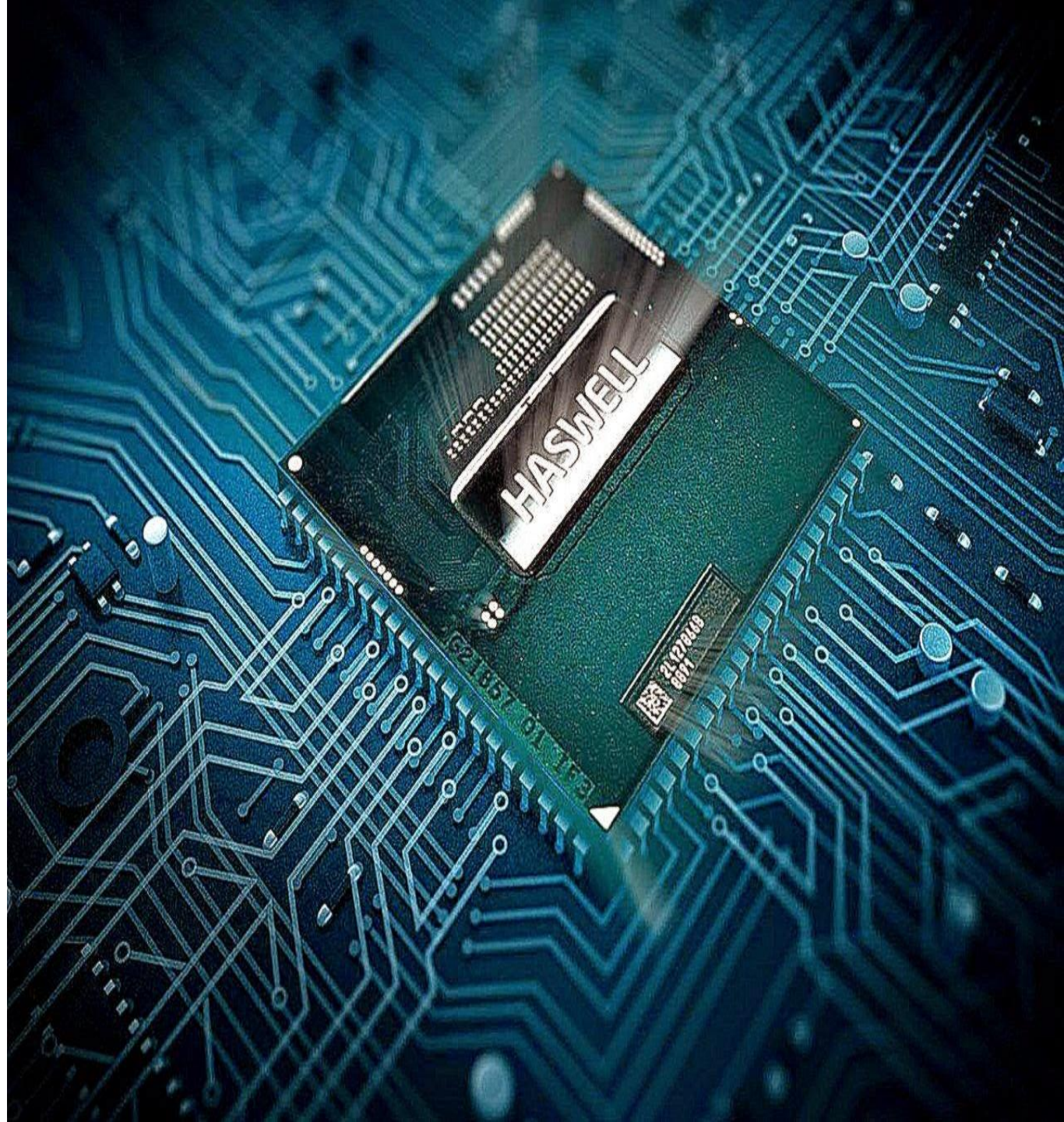
ES 336: Computer Organization and Architecture



Sameer G. Kulkarni
*Assistant Professor,
Department of CSE & EE,
Indian Institute of Technology Gandhinagar*

Today's Lecture

- Computer Organization and Architecture (COA)
 - Course Logistics
- Course Overview
 - What you can expect in this course?
 - What is COA?
 - Why Learn COA?

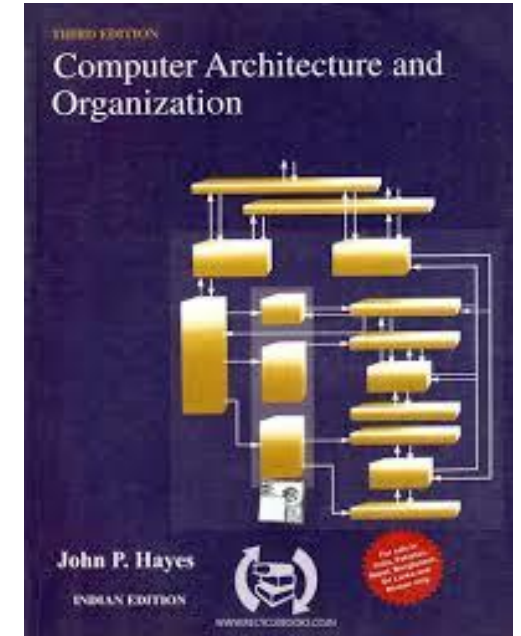
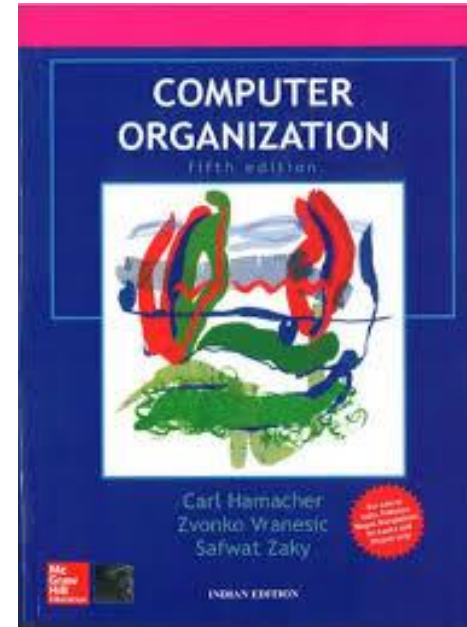
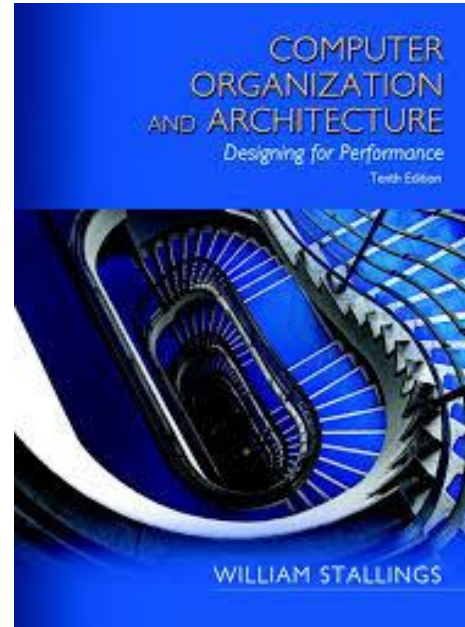
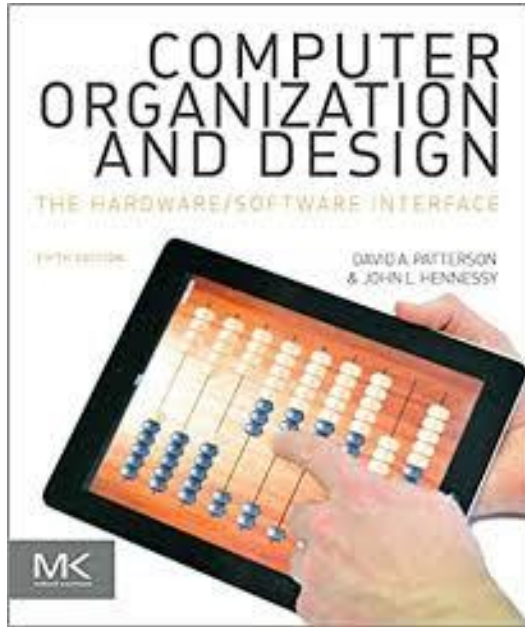


CLASS STRUCTURE AND FORMAT

- Lectures:
 - Tuesday, Thursday 5:00-6:20 PM (*Jasubhai Auditorium*)
- Tutorial sessions:
 - Monday 5:00-6:20 PM **OR** Friday 10:00 -11:20 (11/101 or 07/209)
- Course Website:
 - Microsoft Teams: [Link](#); Team code: **hzd84xg**
- Course Outline: [Google Docs](#) (tentative)
- Contacts:
 - Instructor: Sameer G. Kulkarni (sameergk@iitgn.ac.in)
 - Tutors: Prof. Joyce Mekie, Sameer G. Kulkarni
 - TAs: Naveen Tiwari, Mallika Chauhan, Parth Dangi, Shubham Jaswal, Shivanshu Kant Prasad, Vishwas Misra, Yasir Mohi Ud Din, Yogendra Kumar.
 - Office Hours: Flexible - After Class Hours; Any other time, you can setup via IM/email.



BOOKS AND MATERIALS USED FOR THIS COURSE



**Special Thanks to Professor Joycee Mekie
for Course related discussions and Slide Materials!**

Course presentation slides are adopted from the materials from the above books and the teachings of Professor Rajesh Balasubramoniam (University of Utah) and Professor SmrutiRanjan Sarangi (IITD).

WHAT YOU WILL LEARN IN THIS COURSE

Computer System Components: CPU, Memory, Interconnection structures and bus interconnections; Programmer Abstractions (ISA), Hardware details. VonNeumann&HarvardArchitecture; Performance Metrics and Measurements.

Machine Instructions: Machine instruction characteristics and types; Assembly language; Memory addressing modes; Register organisation; Instruction format; Instruction cycle and instruction pipelining; Reduced instruction set computer (RISC) architecture.

Central Processing Unit: CPU structures and functions; Boolean algebra and gate networks; Moore's Law; Arithmetic and logic unit (ALU). Pipelining – Basic and Advanced; Parallelism (ILP, DLP, TLP); Amdahl's Law; Little's Law; Data and Control Hazards, mechanisms to alleviate them.

Memory Systems: Overview of memory systems; Semiconductor and cache memory; Memory hierarchy design principles and tradeoffs, Cache Coherence, Memory Consistency Models.

WHAT YOU WILL LEARN IN THIS COURSE

IF time permits:

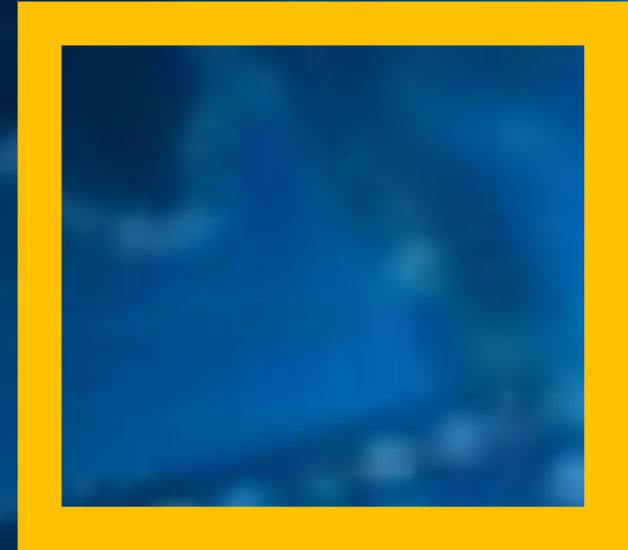
Storage devices: Hard drives CD/DVDs; Evolution of disk storage (SATA/SSD); RAID levels; NAS and SAN;

I/O (External) devices: Input/Output (I/O) modules; Programmed and interrupt I/O; Direct memory access (DMA) and I/O channels;

Massively Parallel Hardware (GPGPUs): Introduction to General Purpose Graphical Processor Units; Architecture, Register File; SMs, Threads.

~~Data Representation and Arithmetic: Integer representation and arithmetic; Floating point and floating point arithmetic.~~

COMPUTER ORGANIZATION AND ARCHITECTURE (COA)



WHAT IS A COMPUTER?

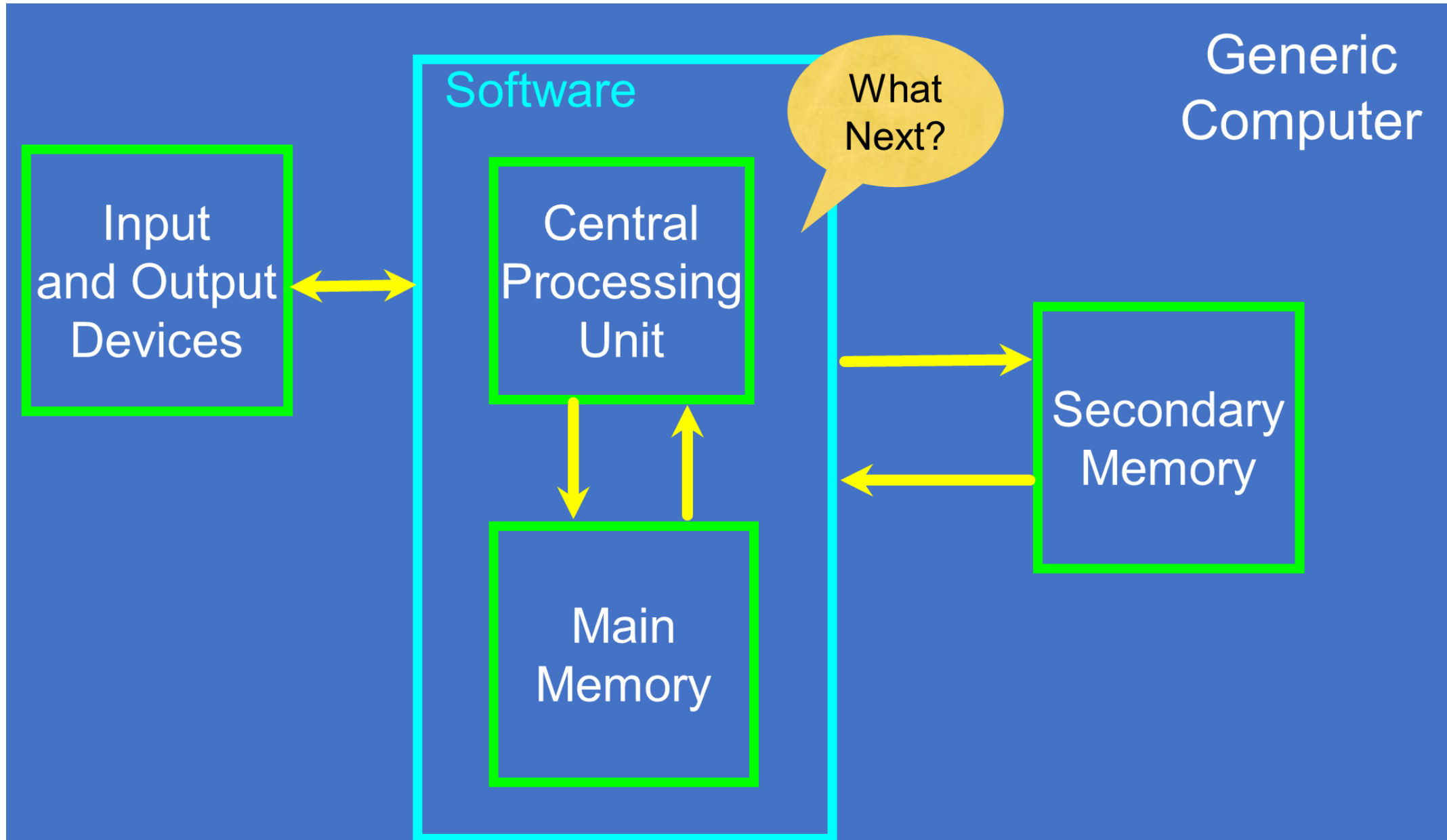


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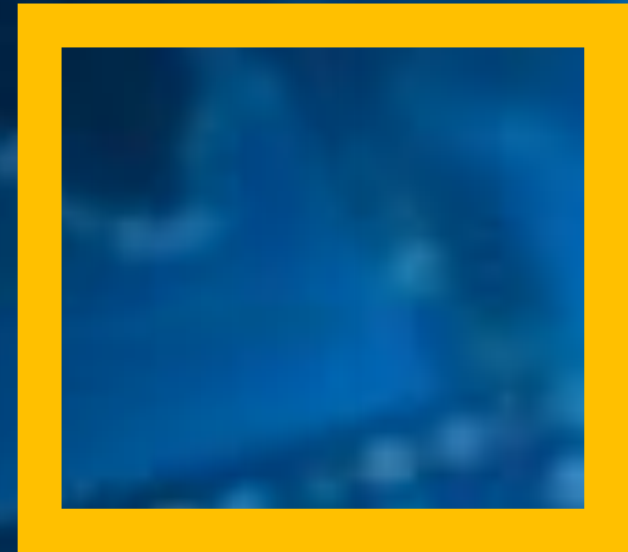
A Digital device that can process Information; can be programmed!!



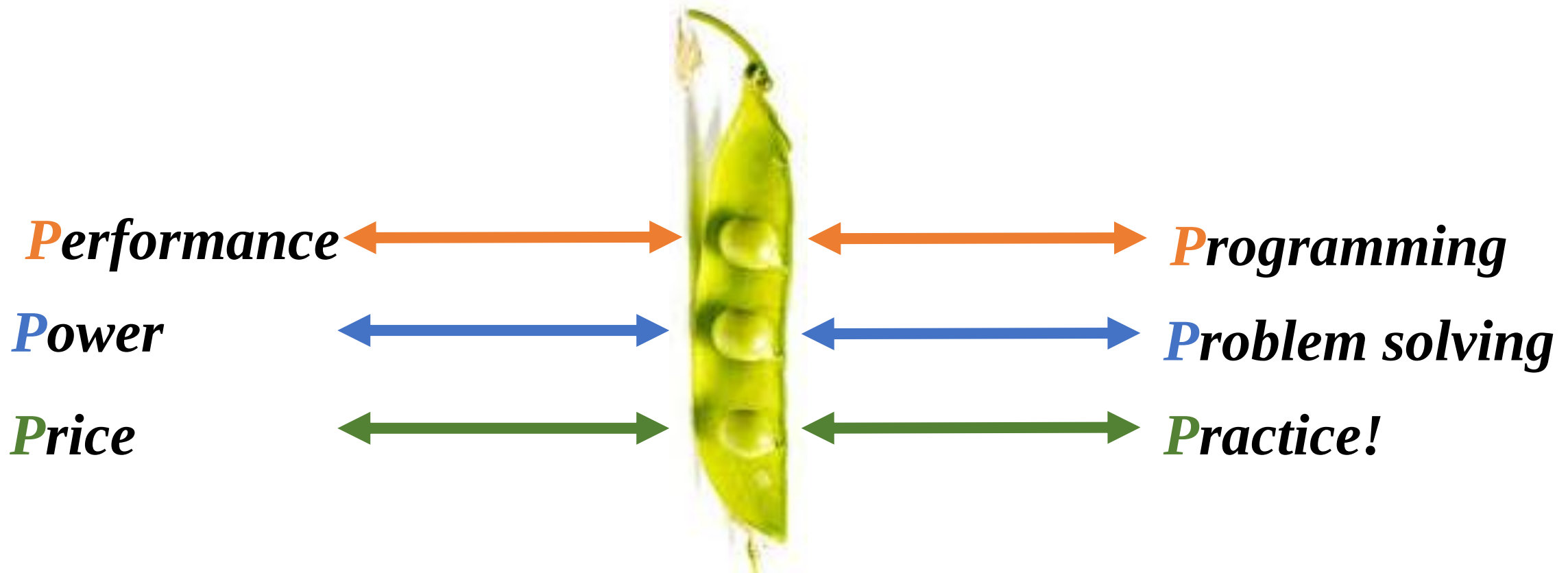
HOW DOES A PROGRAM EXECUTE?



WHY LEARN COMPUTER ORGANIZATION AND ARCHITECTURE?



- *Get deeper understanding of the 3Ps*

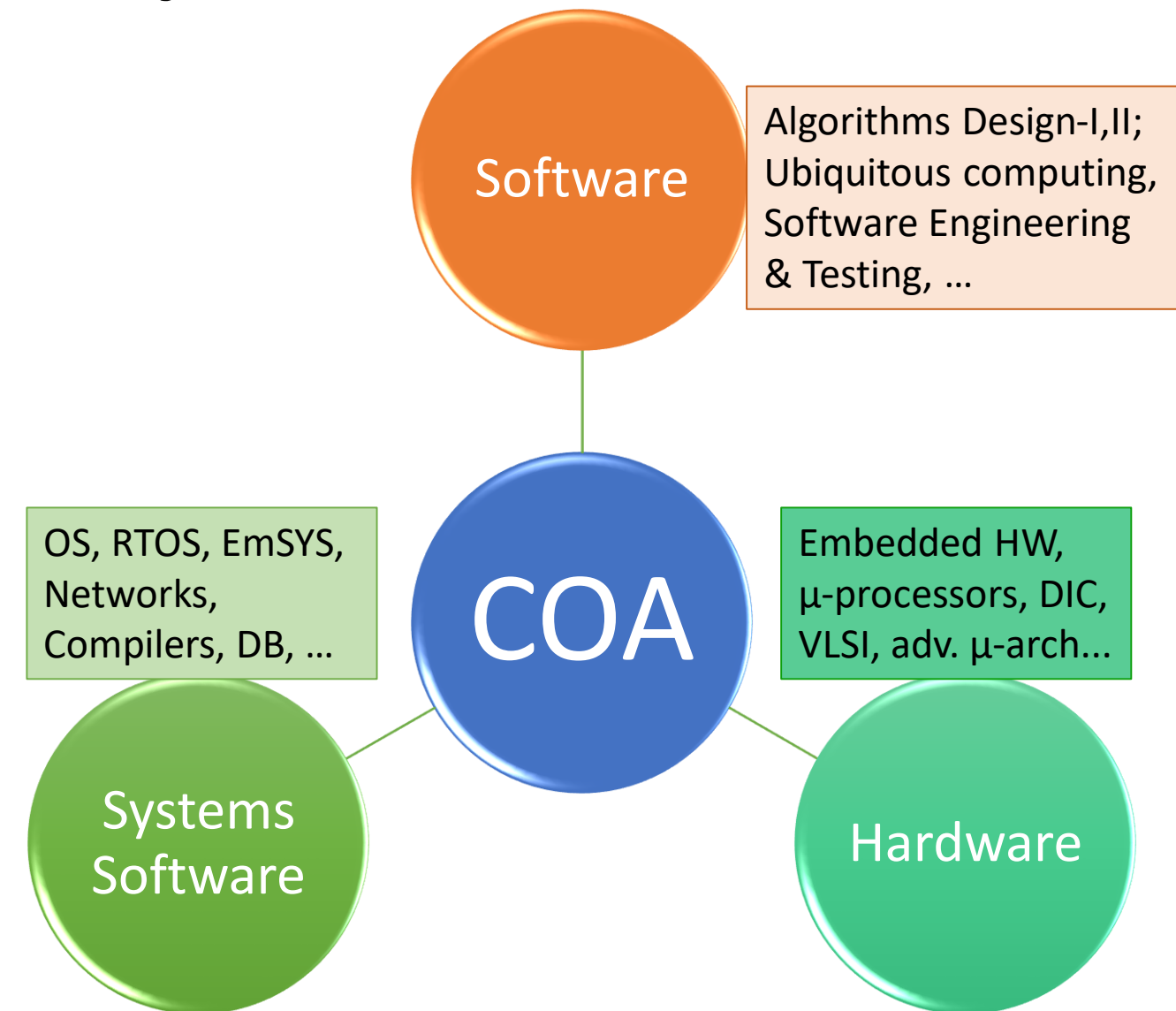


- Make better choice on what hardware you want to invest your money on.
- Understand what it means to benchmark a system, space-time analysis.
- Build your own Computers!

- ***Get deeper understanding of the 3Ps : Performance, Power and Price tradeoffs.***
 - Make better choice on what hardware you want to invest your money on.
 - Understand what it means to optimize, benchmark a system, space-time tradeoff analysis.
- ***Build better and efficient computing systems***
 - Design better Hardware systems
 - Optimize Software (program behavior) for the Hardware
- ***Get Hired by Google, Facebook, Intel, Nvidia, AMD, Qualcomm...***
- ***Build a good repertoire of knowledge to be a Systems Researcher!***

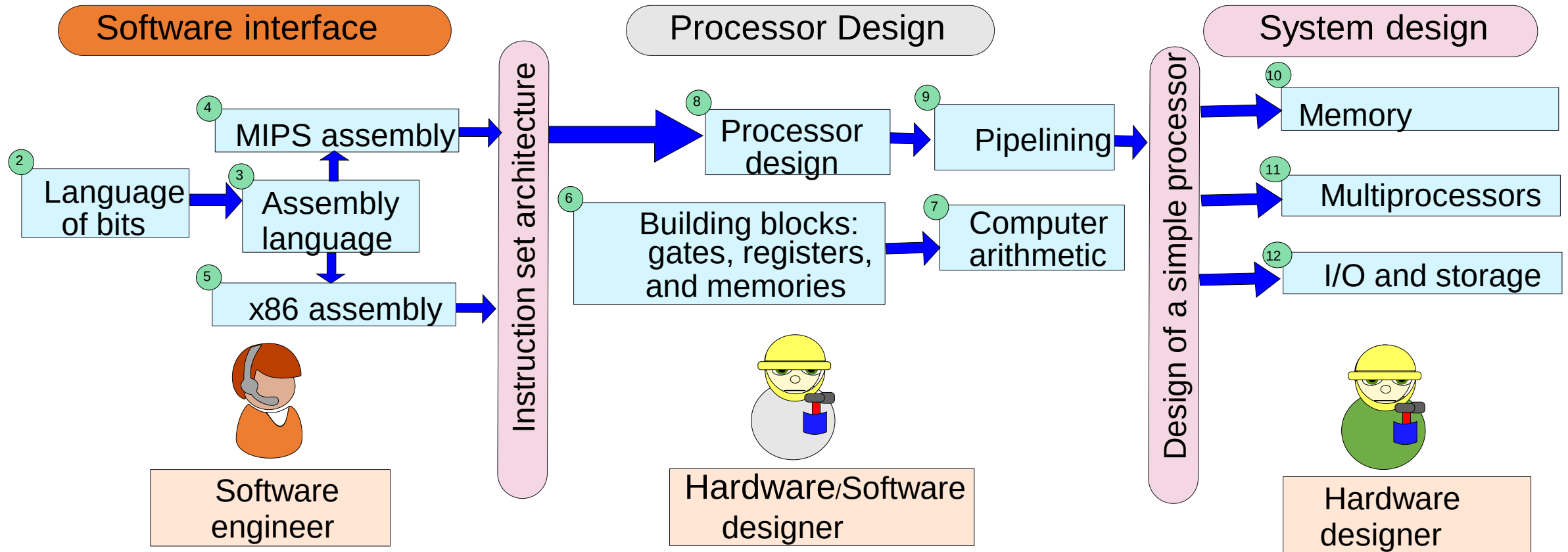


- **Why is COA a core course?**



- **Digital Systems design/Digital Logic**
 - *is the foundational course and a soft prerequisite to learn COA.*
- **Several CS/EE courses build on top of COA.**
 - *A necessary prerequisite to better understand the software and hardware aspects of computing systems.*
 - *GPGPUs, Tensor Processors also need a sound knowledge of COA.*

PERSPECTIVES AND PROSPECTS OF LEARNING COA



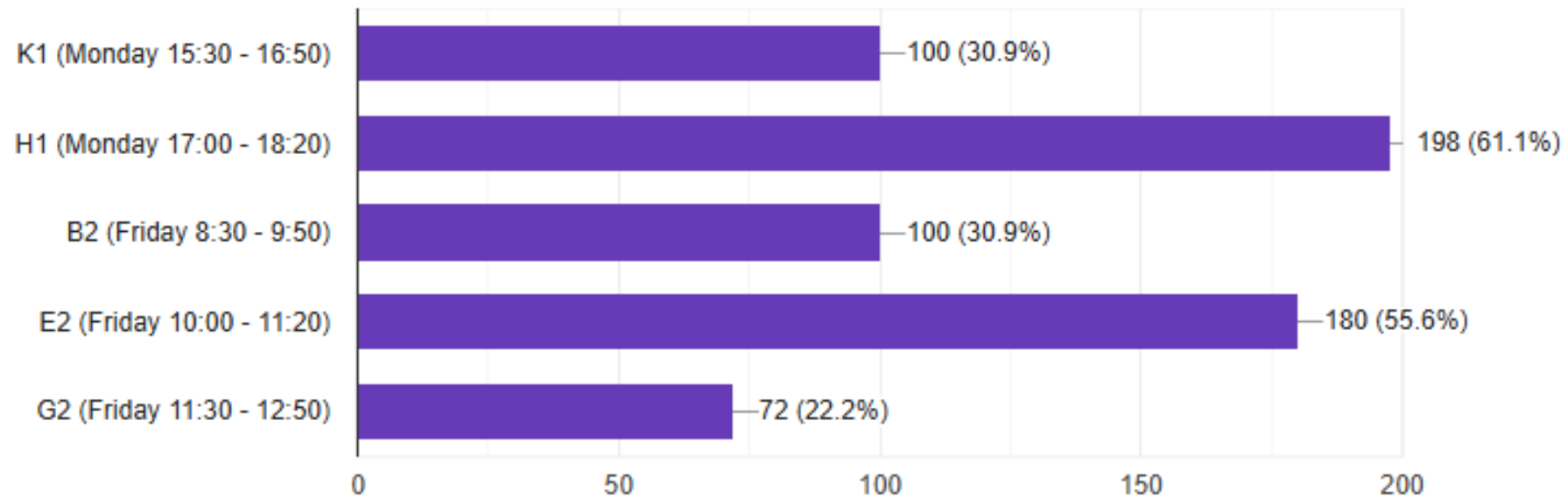
This picture is from COA by Prof. Smruti Ranjan Sarangi.

TUTORIAL SLOT RESPONSE

Which of these time slots can you attend? Select all available slots.

 [Copy chart](#)

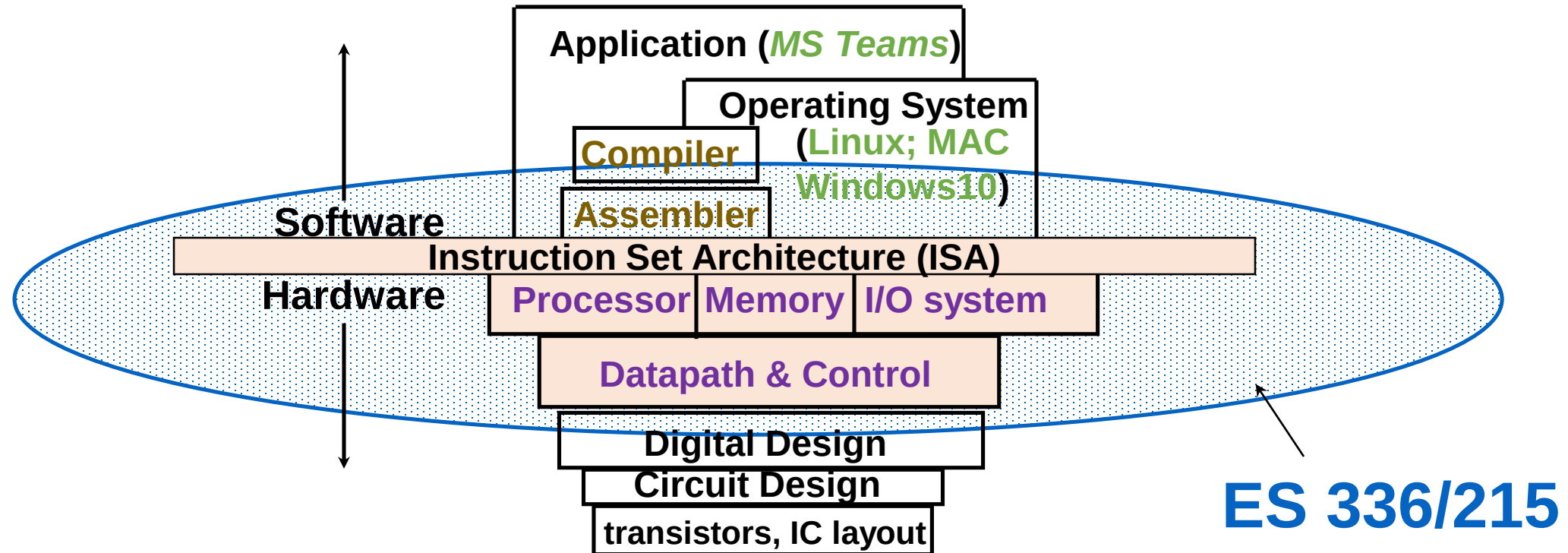
324 responses



OBJECTIVES FOR TODAY

- Understand the working of a software program.
 - Birds eye view of study on COA – Black Box; Grey Box and White Box.
 - High Level Language (Machine Independent – Human Readable)
 - Assembly Language (Machine Dependent – Human Readable)
 - Machine Language (Machine Dependent – Binary, not easily interpretable).
- The eight great ideas that have shaped the COA.
 - Key Ideas behind COA; -- Highlight; *some intentionally skipped*;
- Performance Metrics.
 - What do we mean by performance?
 - How can we quantify the performance and compare two different hardware/programs?
- Textbook material:
 - Parts of Chapter 1.1, 1.2 (COAHP)
 - Parts of Chapter 1 (COAWS)

COA: BIRD'S EYE VIEW

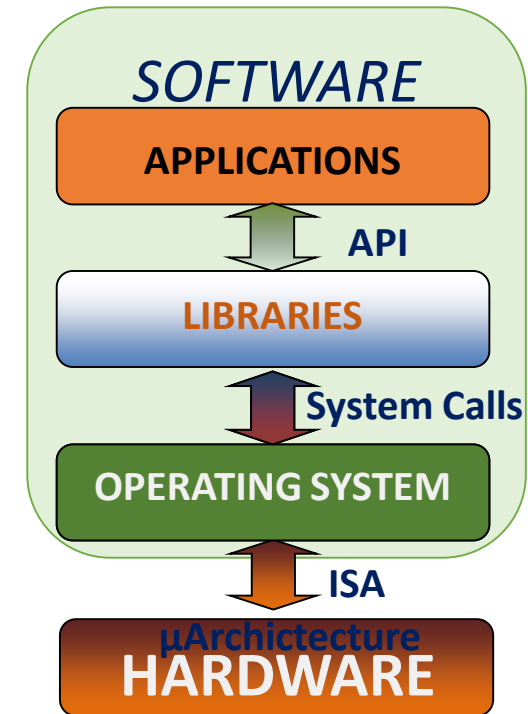


- Key Idea: *levels of abstraction*
 - hide unnecessary implementation details
 - helps us cope with enormous complexity of real systems

LAYMAN'S TERMS: WHAT DO WE MEAN BY COA?

- What is a Computer?
 - *A Digital device that can gather, manipulate & store information.*
- What is Architecture?
 - The art and practice of designing;
e.g. Civil Engineer-Architecture (Blueprint)
 - An abstraction/high-level view of the system
 - to whom, what & why?
- What is Organization?
 - Ensemble of closely related entities with common purpose.
 - A representation of group or a system of interconnected things. –
how and why?

Computer Organization *a.k.a.* MicroArchitecture



COMPUTER ARCHITECTURE AND COMPUTER ORGANIZATION

- Attributes of a system that are visible to the programmer.
- Have a direct impact on the logical execution of a program

- Instruction set, number of bits used to represent various data types, I/O mechanisms, techniques for addressing memory

Computer Architecture

Architectural attributes include:

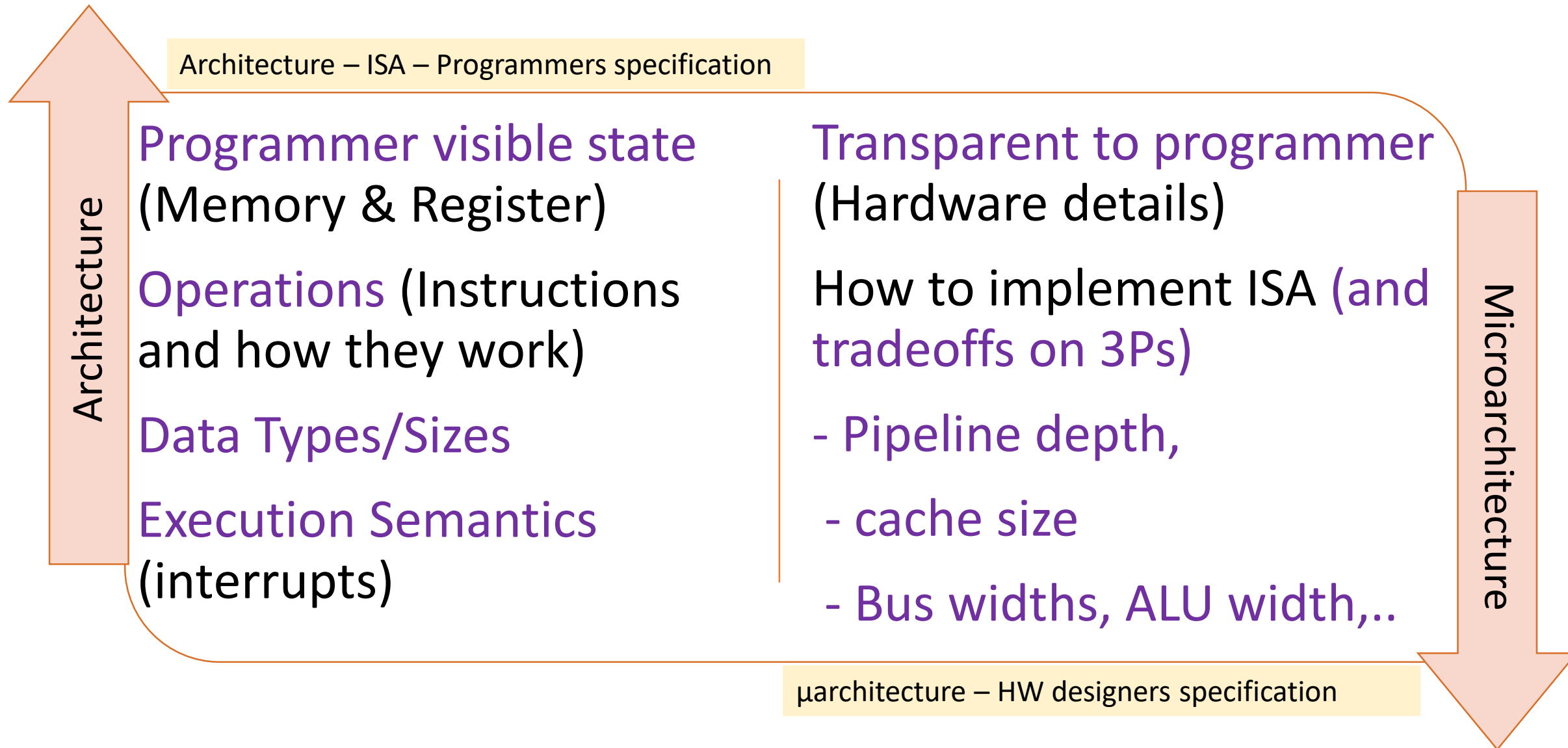
Organizational attributes include:

Computer Organization

- Hardware details transparent to the programmer, control signals, interfaces between the computer and peripherals, memory technology used

- The operational units and their interconnections that realize the architectural specifications

1 SLIDE SUMMARY: ARCHITECTURE VS MICROARCHITECTURE



EIGHT GREAT IDEAS IN COMPUTER ARCHITECTURE

Chap 1.3
(COAHP –ed 5);
Chap1: COAWS

Design for Moore's law

Use Abstraction to Simplify design

Make the Common case fast

Performance Via Parallelism

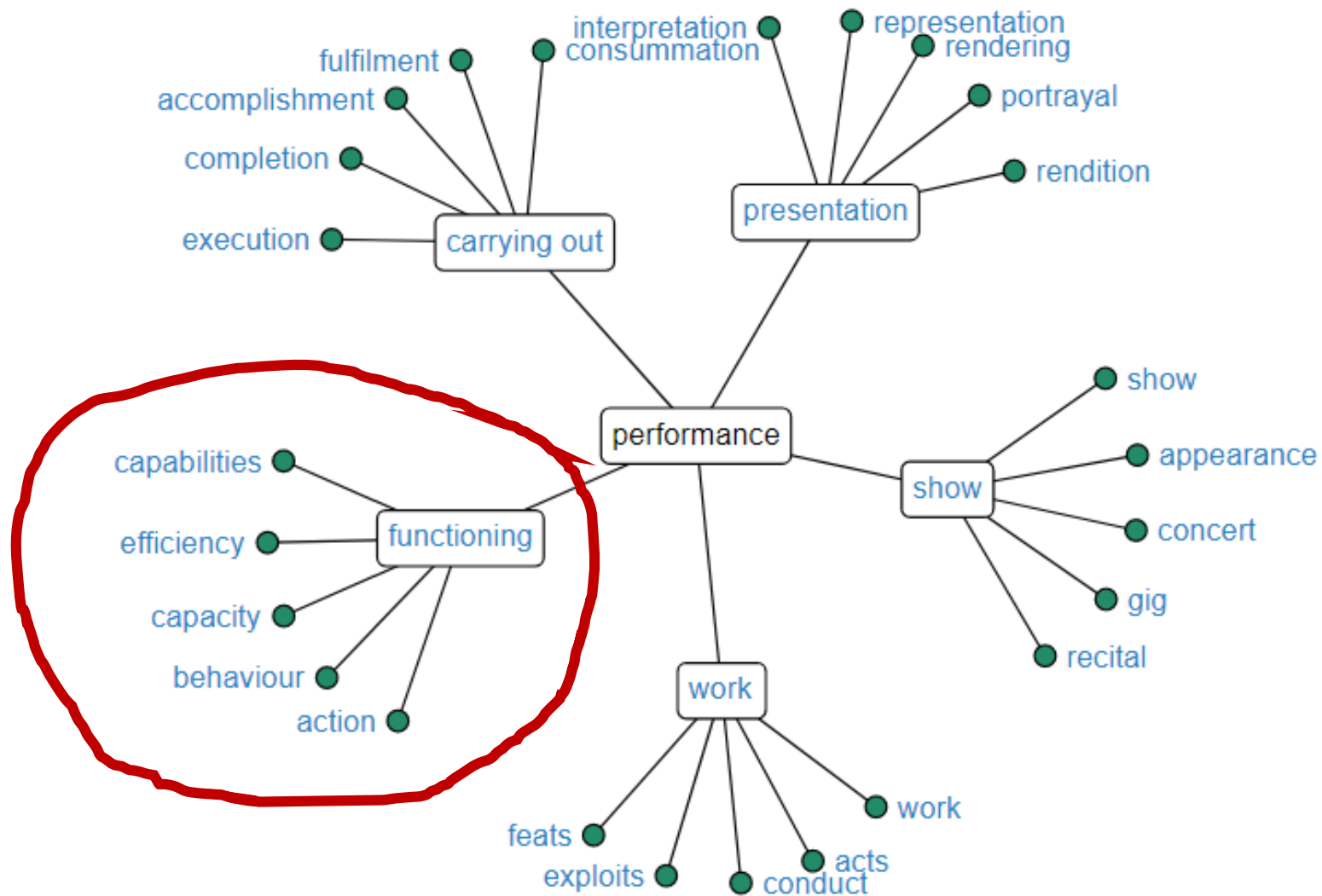
Performance Via Pipelining

Performance Via Prediction

Hierarchy of Memory

Dependability Via Redundancy

PERFORMANCE – WHAT DOES IT MEAN?



This picture is from Thefreedictionary.org --
<https://www.freethesaurus.com/performance>

- Which Vehicle has the best performance?

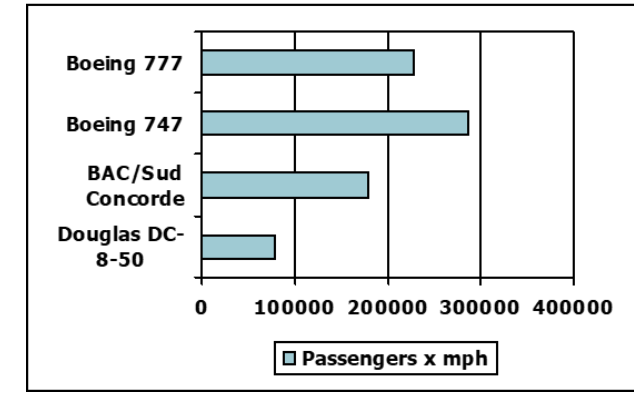
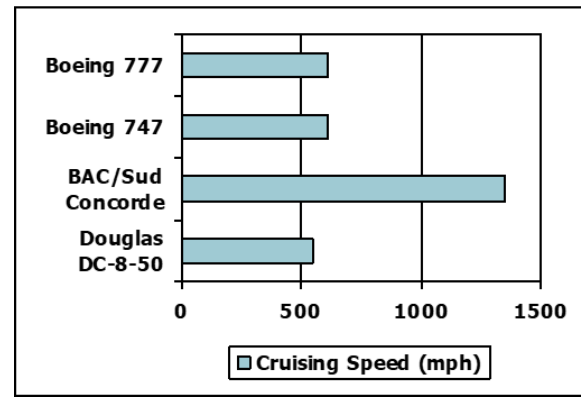
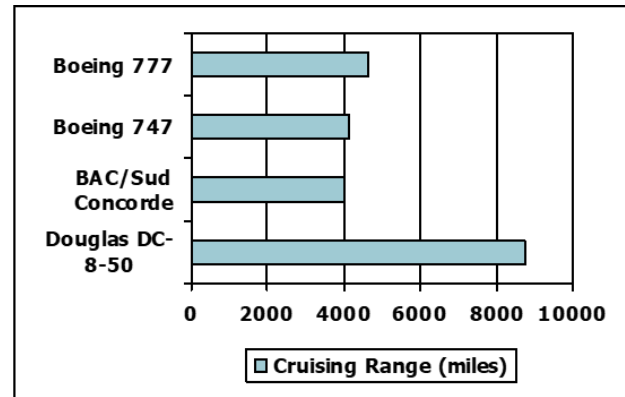
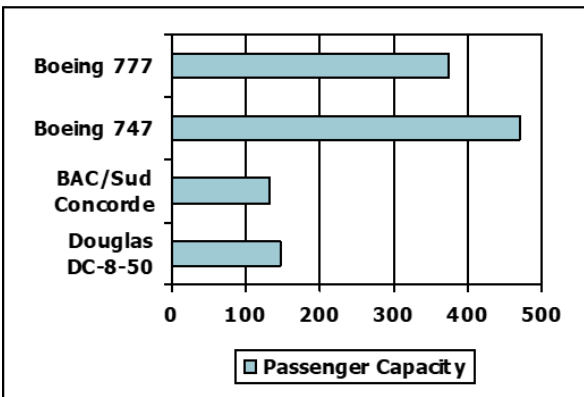
Automobile	Capacity (#persons)	Speed (Kmph)	Mileage (Kmpl)	Time to reach Mumbai (500Kms)	Throughput (persons/hr)	Fuel Efficiency (persons/ltr.)
Bike	2	50	80	10 hrs	0.2 pph	0.32 ppl
Car	5	100	30	5 hrs	1 pph	0.30 ppl
Bus	50	100	20	5 hrs	10 pph	2 ppl

- Time to do a task
 - Execution time/Response time/Latency
- Tasks performed per unit time (per day, hr, seconds)
 - Throughput
- Other Metrics:
 - Efficiency (Energy consumed to per task) etc.

Vehicle Throughput:

The rate at which the persons reach the destination in a given vehicle.

- Which airplane has the best performance?



Airplane	Passenger capacity	Cruising range (miles)	Cruising speed (m.p.h.)	Passenger throughput (passengers x m.p.h.)
Boeing 777	375	4630	610	228,750
Boeing 747	470	4150	610	286,700
BAC/Sud Concorde	132	4000	1350	178,200
Douglas DC-8-50	146	8720	544	79,424

Passenger Throughput:

The rate at which the airplane transports the passengers.

RESPONSE TIME AND THROUGHPUT

- Response time (Latency)
 - How long it takes to do a task.
 - Execution time; Cycles per Instruction (CPI); Avg. Memory access time (AMAT)
- Throughput
 - Total work done per unit time.
 - e.g., tasks/transactions/... per hour
 - Transactions per second (TPS); Million Instructions per second (MIPS)
- How are response time and throughput are affected by
 - Replacing the processor with a faster version?
 - Adding more processors?

RELATIVE PERFORMANCE

- Define Performance = *Time need to do a unit work*. **<smaller the better>**
OR = *Unit of Things done per unit time*. **<Larger the better>**

$$\text{Performance}(x) = \text{Execution time}(x) \text{ seconds}$$

$$\text{Performance}(x) = 1 / \text{Execution time}(x) \text{ seconds}^{(-1)}$$

- “X is n time faster than Y” or “X is n times as fast as Y” or “From Y to X the speedup is n times ”

$$\begin{aligned} \text{Speedup}(x/y) &= \text{Performance}(x) / \text{Performance}(y) \\ &= \text{Execution time}(y) / \text{Execution time}(x) \end{aligned}$$

- Example: time taken to run a program (*same program on different machines*)
 - Program A takes **10s** on Machine **X**, and **15s** on Machine **Y**
 - Execution Time(**Y**) / Execution Time(**X**)
= 15s/10s = 1.5
 - So **X** is 1.5 times faster than **Y**; **Speedup(X/Y) = 1.5**